

## SCHOOL OF ENGINEERING, CUSAT

B. TECH DEGREE 2<sup>nd</sup> SEMESTER SECOND INTERNAL EXAMINATIONS, JULY 2024

## 23-200-0205B INTRODUCTION TO CYBER PHYSICAL SYSTEMS

(CSA, CSB, EEA, EEB, ITA, ITB, ECA, ECB)

(2023 Scheme)

TIME: 2 hrs

MAX. MARKS: 30

## Course Outcomes

On completion of this course the student will be able to:

|     |                                                                                                                |
|-----|----------------------------------------------------------------------------------------------------------------|
| CO1 | Understand the features & components of Cyber Physical Systems                                                 |
| CO2 | Understand the elementary constructs of Arduino Software                                                       |
| CO3 | Develop optimal programs & circuits for interfacing various sensors with Arduino                               |
| CO4 | Apply Arduino IDE for developing suitable programs for interfacing sensors & actuators with Arduino & Node MCU |

## PART A

(5 x 2= 10 Marks)

(Answer all questions)

| Q.No. | Questions                                                                | Marks | BL | CO | PO |
|-------|--------------------------------------------------------------------------|-------|----|----|----|
| I (a) | What functions are used to setup serial communication in Arduino?        | 2     | L3 | 2  | 1  |
| (b)   | How does the analogRead() function work in Arduino?                      | 2     | L2 | 2  | 1  |
| (c)   | How can you configure a pin as an input or output in Arduino?            | 2     | L3 | 2  | 3  |
| (d)   | What function is used to connect NodeMCU to a WiFi network?              | 2     | L3 | 4  | 2  |
| (e)   | What is a pull-up resistor, and how is it used in a push button circuit? | 2     | L2 | 3  | 1  |

B

**PART B**  
( Answer the following questions  
choosing either the first OR second option)

| (1 x 10 = 10 Marks)        |                                                                                 |       |    |    |    |
|----------------------------|---------------------------------------------------------------------------------|-------|----|----|----|
| Q No.                      | Questions                                                                       | Marks | BL | CO | PO |
| II (a)                     | What is PWM, and how is it implemented in Arduino?                              | 6     | L2 | 2  | 5  |
| (b)                        | Explain the difference between sinking and sourcing methods in LED interfacing. | 4     | L4 | 2  | 3  |
| <b>OR</b>                  |                                                                                 |       |    |    |    |
| III (a)                    | How can you display a value on an LED bar graph using Arduino?                  | 5     | L6 | 3  | 1  |
| (b)                        | Explain how to interface an analog temperature sensor with Arduino.             | 5     | L3 | 1  | 1  |
| <b>(1 x 10 = 10 Marks)</b> |                                                                                 |       |    |    |    |
| IV (a)                     | List some key features of the NodeMCU/ESP32                                     | 4     | L1 | 4  | 5  |
| (b)                        | Explain the interfacing between any motor (DC/Stepper/Servo) with Arduino.      | 6     | L3 | 4  | 3  |
| <b>OR</b>                  |                                                                                 |       |    |    |    |
| V (a)                      | Write a simple code to blink an LED connected to pin D1 on NodeMCU.             | 5     | L3 | 4  | 1  |
| (b)                        | How do you connect NodeMCU to a WiFi network?                                   | 5     | L3 | 4  | 6  |

**Bloom's Taxonomy Levels**

Bloom's taxonomy levels (L1-Remember, L2-Understand, L3-Apply, L4-Analyze L5-Evaluate L6-Create), PO-Program outcome

| L1-% | L2- % | L3-% | L4-% | L5-% | L6-% |
|------|-------|------|------|------|------|
| 7    | 23    | 53   | 7    | 0    | 7    |